

ACH Batch Payment Processing Failure - Year-End Volume Spike

Meridian Commercial Bank - Internal Document

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Summary

The nightly ACH batch processing job on 2025-12-19 failed partway through execution, leaving approximately 6,300 outbound ACH payments (payroll and vendor disbursements) unprocessed and delayed by one business day.

Timeline

- 02:00 - Nightly ACH batch job starts (volume: 118,000 transactions, about 2.4x normal due to year-end payroll runs).
- 02:41 - Batch worker pool begins reporting OutOfMemoryError on 3 of 12 worker nodes.
- 02:44 - Job orchestrator marks batch as FAILED and halts remaining shards to avoid partial/duplicate submission to the Fed ACH network.
- 03:15 - Payments on-call paged via automated batch-failure alert.
- 07:30 - Engineering team (business hours start) confirms 6,300 of the 118,000 transactions in the failed shards were not submitted.
- 09:00 - Manual re-run of the 3 failed shards after increasing worker memory limits.
- 10:12 - Remaining transactions submitted successfully to the Fed ACH network same-day, but after the 09:00 same-day submission cutoff, so they settled one business day late.

Root Cause

ACH batch workers were provisioned with a fixed memory limit sized for average daily volume (about 50,000 transactions). The year-end payroll spike (118,000 transactions) was not accounted for in capacity planning, causing in-memory batch buffers on 3 worker nodes to exceed the container memory limit.

Impact

- 6,300 payroll/vendor ACH payments delayed by one business day.
- No payments were lost or duplicated; the halt-on-failure design worked as intended to prevent double submission.
- Customer support fielded approximately 140 inquiries from affected payees.

Resolution

- Re-ran failed shards with increased memory limits.
- Notified affected business customers of the one-day delay.

Action Items

1. Add seasonal/calendar-aware capacity planning for ACH batch workers, particularly around month-end and year-end payroll cycles (Owner: Payments Platform, Due: 2026-01-31).
2. Implement dynamic worker autoscaling based on queued transaction count rather than a fixed worker pool (Owner: Payments Platform, Due: 2026-02-28).
3. Add a pre-flight volume estimate check that pages on-call proactively when a batch run is more than 1.5x the trailing 30-day average (Owner: SRE, Due: 2026-01-15).

Related Incidents

This is the third payments-related incident in Q3-Q4 2025 involving resource exhaustion under load (see INC-2025-08-14-001 connection pool exhaustion, INC-2025-10-02-002 database write load). Recurring theme: capacity limits not scaled to real-world peak/seasonal load.